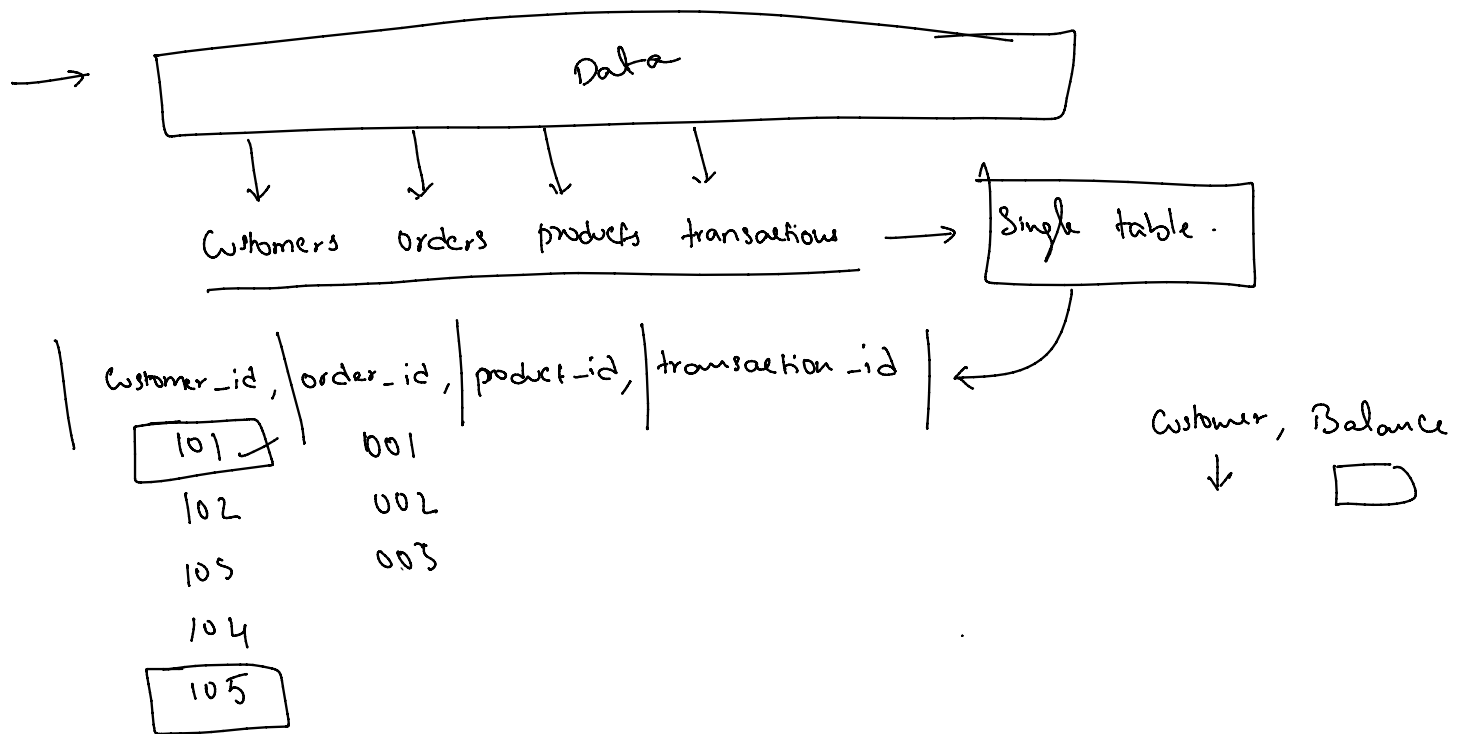


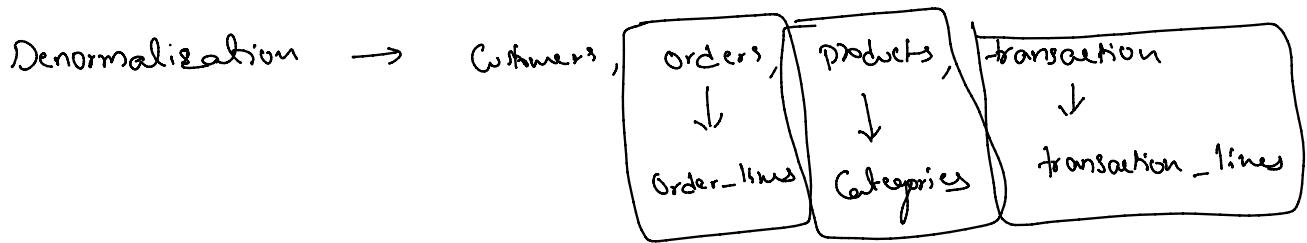
What is Data & why do we need it?

- * Data is a information Collected to understand the nature of Substance.
- * When we understand the nature of substance, we understand how to use it.

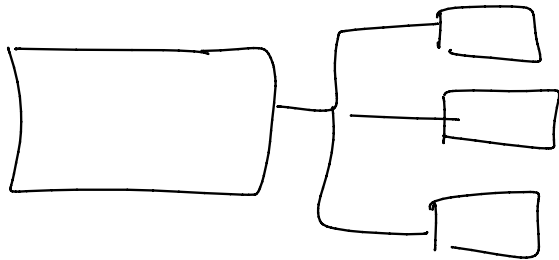


Normalization → Dividing data into multiple tables keep logical Relationship between the tables.

- * Removing of data Redundancy
- * Data integrity
- * Easily we can extract the data from a table.



Denormalization is the process of combining the data into a single table from multiple tables.



Normal Forms → 6 types of Normal Forms

1st Normal Form →

Col	Col	Col	Col
101	A	B	C
101	A	C	B

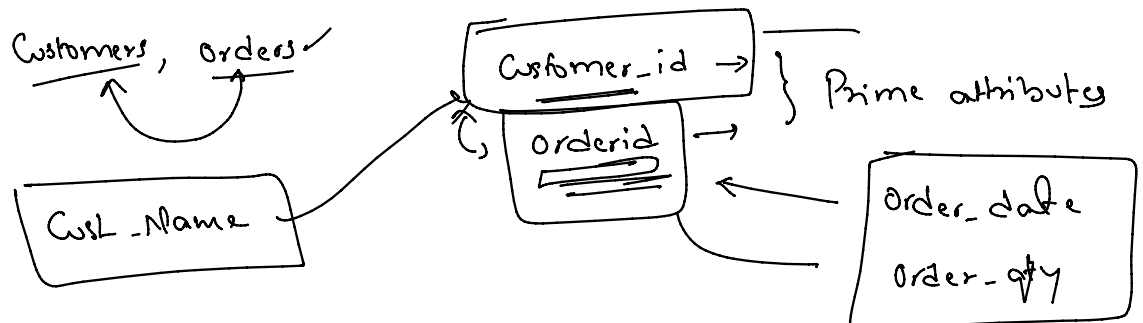
→ Unique
→ Unique

Cell → a Cell holds a value

Column → a Column has multiple Cells.

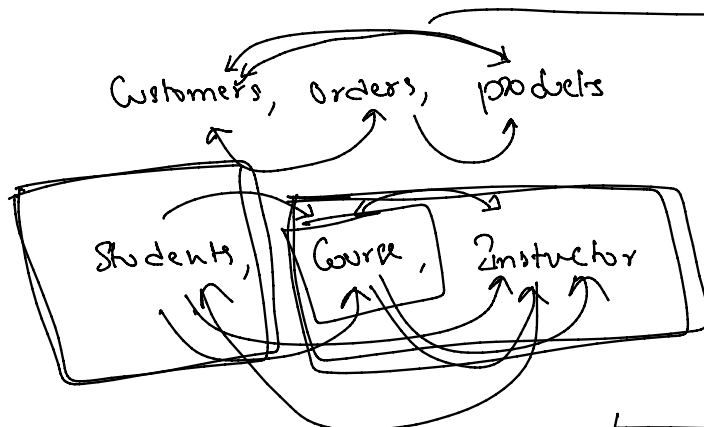
Row → a Row is formed with Columns.

2nd Normal Form → 1NF + additional rules (no partial dependency)

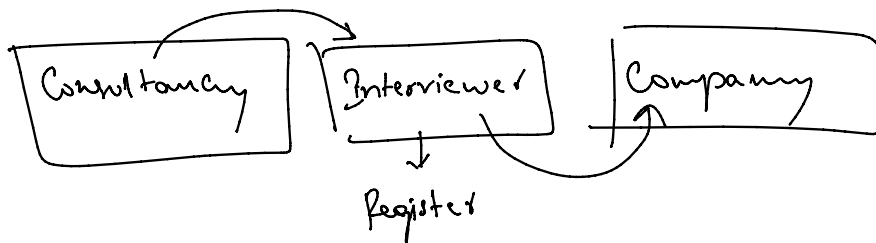
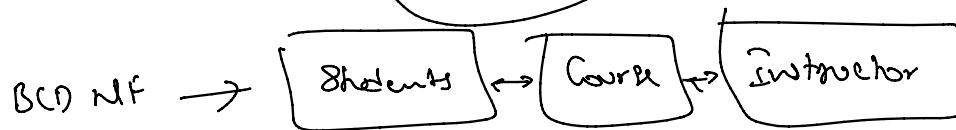


... additional rules (no transitive dependency)

3NF → 2NF + additional rules (no transitive dependency)

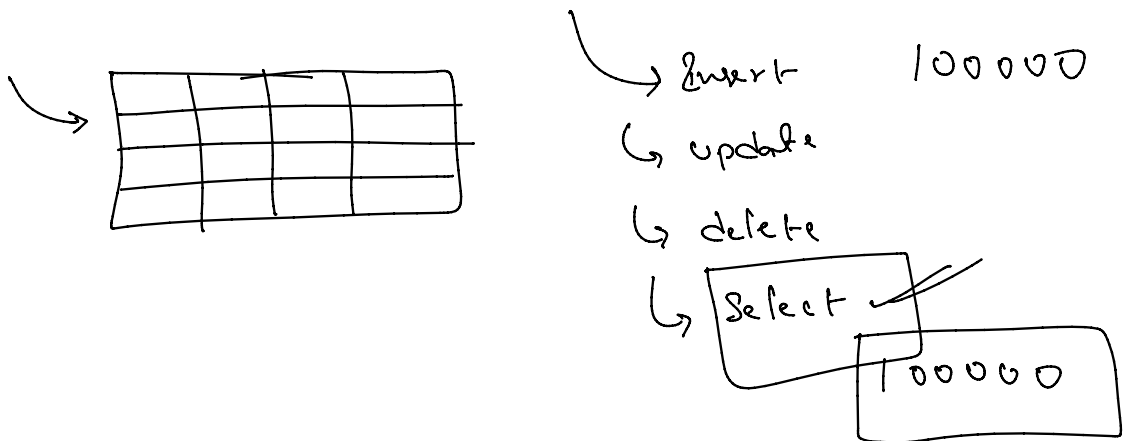


Students → Course → Instructor.



Types of Systems in DBMS

OLTP → Online Transactional Processing (Transactional tables)



OLAP → Online Analytical Processing (warehouse) tables

relation

Denormalization

